

```

rm(list = ls())

# Local github
codepath <- c("/media/larryleon/My Projects/GitHub/Forest-Search/R/")
source(paste0(codepath, "source_forestsearch_v0.R"))
source_fs_functions(file_loc = codepath)

library(kableExtra)
library(knitr)
library(ggplot2)
library(gridExtra)
library(cubature)
library(aVirtualTwins)
library(randomForest)
library(survival)
library(survminer)
library(grf)
library(policytree)
library(data.table)
library(plyr)
library(dplyr)
library(glmnet)
library(corrplot)

library(table1)

maxFollow <- 84
cens.type <- "weibull"

# m1 -censoring adjustment
muC.adj <- log(1.5)
k.z3 <- 1
k.treat <- 0.9

z1_frac <- 0.25 # Default model index 'm1' (The 1st quartile of z1=er)
pH_super <- 0.125 # non-NULL re-defines z1_frac

if (is.null(pH_super)) {
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_frac)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_frac)))
  cat("Underlying pH_super", c(pH_check), "\n")
}
# pH_super specified If pH_super then override z1_frac and find z1_frac to
# yield pH_super

if (!is.null(pH_super)) {
  # Approximate Z1 quantile to yield pH proportion
  z1_q <- uniroot(propH.obj4, c(0, 1), tol = 1e-04, pH.target = pH_super)$root
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_q)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_q)))
  cat("pH", c(pH_check), "\n")
  rel_error <- (pH_super - pH_check)/pH_super
  if (abs(rel_error) >= 0.1)

```

```

    stop("pH_super approximation relative error exceeds 10%")
    z1_frac <- z1_q
    cat("Underlying pH_super", c(pH_check), "\n")
  }

## pH 0.122449
## Underlying pH_super 0.122449

# Bootstrap on log(hr) scale converted to HR (est.loghr=TRUE & est.scale='hr')
t.start.all <- proc.time()[3]

##### Forest search criteria

hr.threshold <- 1.25 # Initital candidates
hr.consistency <- 1 # Candidates for many splits
pconsistency.threshold <- 0.9
stop.threshold <- 0.95
maxk <- 2
nmin.fs <- 60
pstop_futile <- 0.5
# Limit timing for forestsearch
max.minutes <- 3
m1.threshold <- Inf # Turning this off (Default)
# pconsistency.threshold<-0.70 # Minimum threshold (will choose max among
# subgroups satisfying)
fs.splits <- 400 # How many times to split for consistency
# vi is % factor is selected in cross-validation --> higher more important
vi.grf.min <- 0.2
# Null, turns off grf screening
d.min <- 10 # Min number of events for both arms (d0.min=d1.min=d.min)
# default=5 Virtual twins analysis Counter-factual difference (C-E) >=
# vt.threshold Large values in favor of C (control)
vt.threshold <- 0.225 # For VT delta
treat.threshold <- 0

maxdepth <- 2
n.min <- 60
ntree <- 1000
# GRF criteria
dmin.grf <- 12 # For GRF delta
# Note: For CRT this represents dmin.grf/2 RMS for control (-dmin.grf/2 for
# treatment)
frac.tau <- 0.6

outcome.name <- c("y.sim")
event.name <- c("event.sim")
id.name <- c("id")
treat.name <- c("treat")

cox.formula.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat"))
cox.formula.adj.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat+v1+v2+v3+v4+v5"))

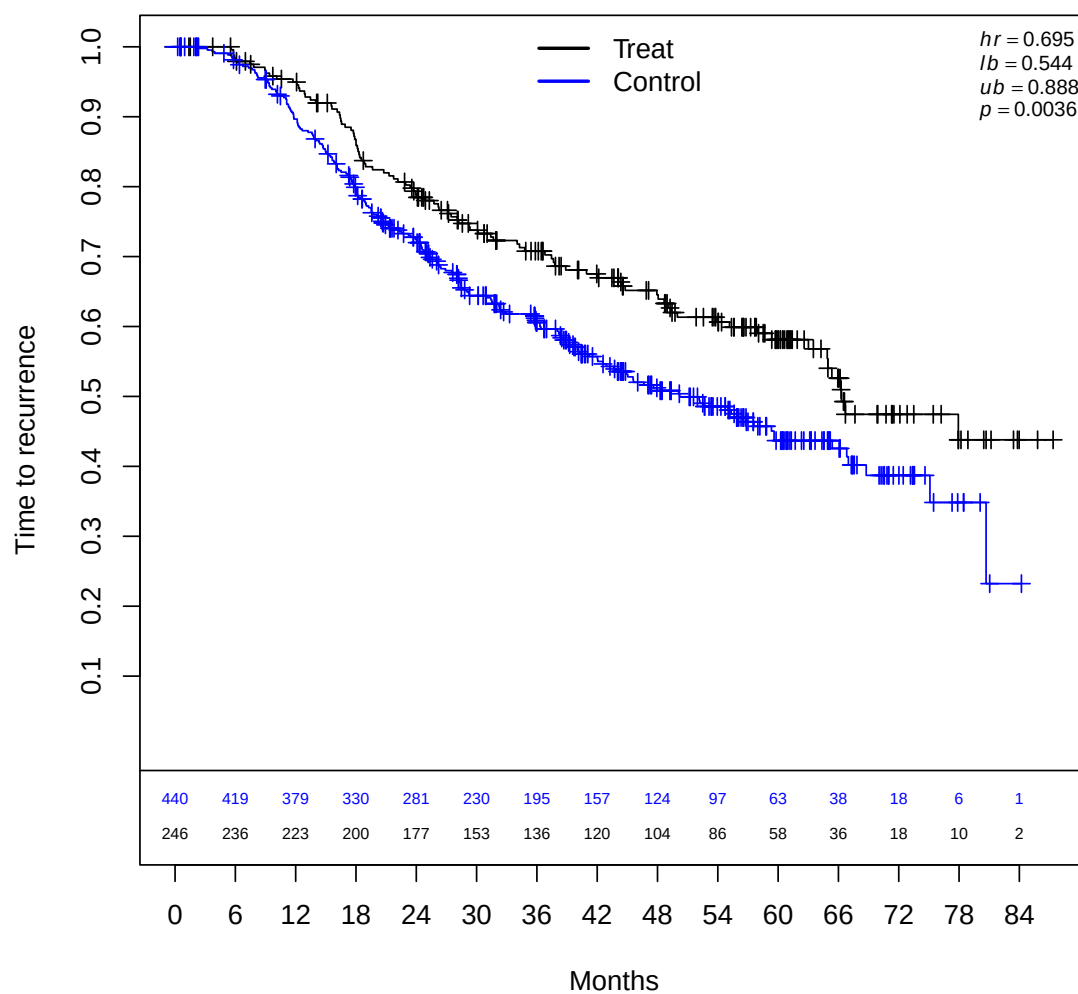
mod.harm <- "alt"
hrH.target <- 2
# out.loc = NULL turns off file creation

```

```

N <- 1000
this.dgm <- get.dgm4(mod.harm = mod.harm, N = N, k.treat = k.treat, hrH.target = hrH.target,
  cens.type = cens.type, out.loc = NULL, details = TRUE, parms_torand = FALSE)

```



```

## Super-population empirical harm and non-harm hazard ratios= 2.000007 0.6466405
## Causal HR (empirical ITT)= 0.7057463

file_out <- c("output/sim-FS4_N1k_hrH=2_sim257.Rdata")

```

```

# sim sensH ppvH sensHc ppvHc lasso_omit 1: 78 0.4402985 0.5412844 0.9422633
# 0.9158249 1 2: 164 0.6146789 0.5275591 0.9326599 0.9518900 1 3: 221 0.3149606
# 0.5063291 0.9553265 0.9055375 1 4: 257 0.5528455 0.5666667 0.9407070
# 0.9375000 1 5: 273 0.5422535 0.5065789 0.9125874 0.9233491 1

```

```
n_add_noise <- 0
```

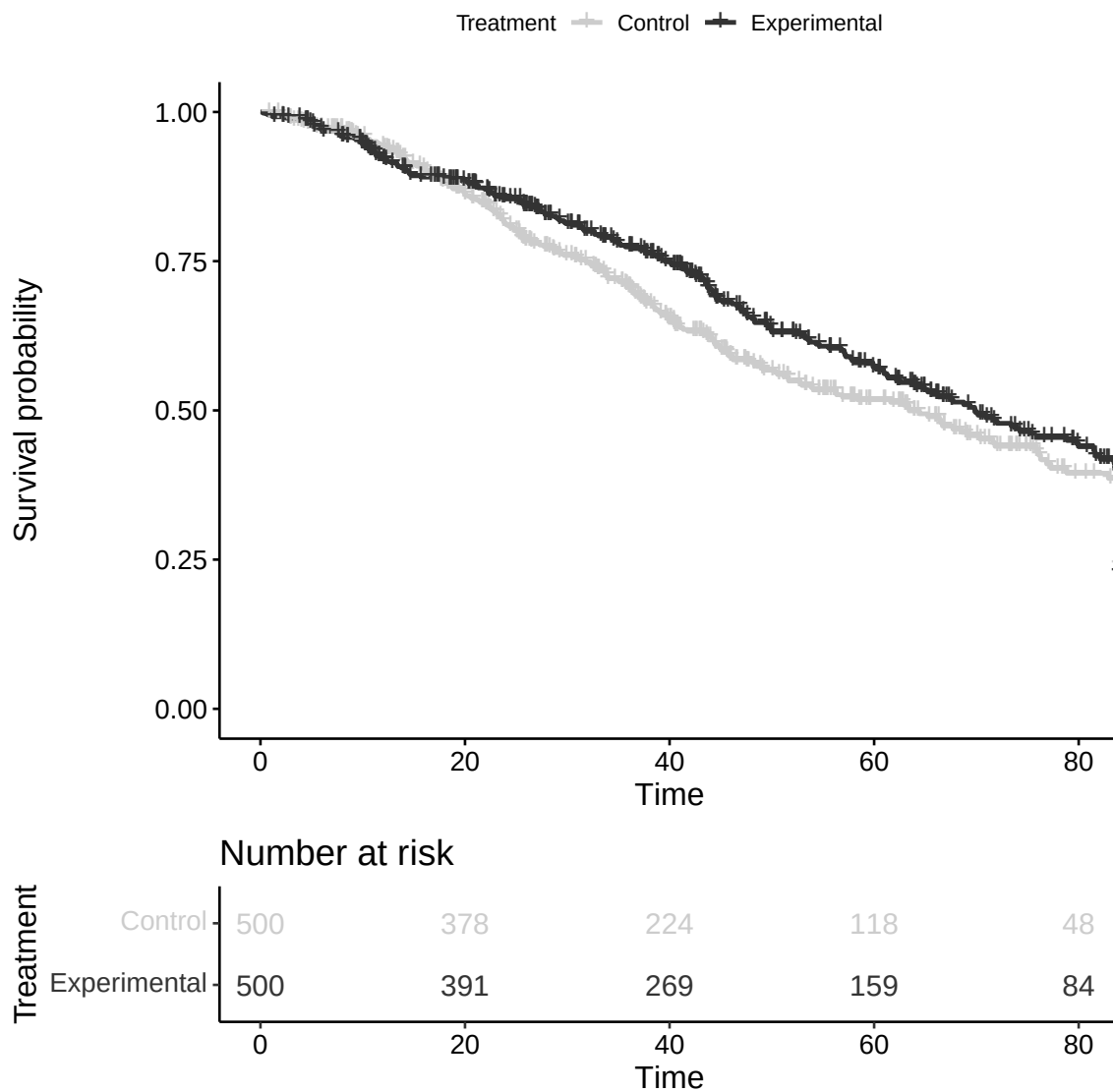
```
sim <- 257
```

```
dgm <- this.dgm$dgm

x <- sim_aftm4_gbsg(dgm = dgm, n = N, maxFollow = maxFollow, muC.adj = muC.adj, simid = sim)

kmfit <- survfit(Surv(y.sim, event) ~ treat, data = x)

ggsurvplot(kmfit, data = x, main = "K-M curves for simulated data", legend = "top",
  legend.title = "Treatment", legend.labs = c("Control", "Experimental"), palette = "grey",
  risk.table = TRUE, risk.table.col = "strata")
```



```
if (n_add_noise == 0) {
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade3")
}

if (n_add_noise == 5) {
  set.seed(8316951 + 1000 * sim)
  # Add 5 noise
  x$noise1 <- rnorm(N)
```

```

x$noise2 <- rnorm(N)
x$noise3 <- rnorm(N)
x$noise4 <- rnorm(N)
x$noise5 <- rnorm(N)
confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade3", "noise1",
  "noise2", "noise3", "noise4", "noise5")
}

if (n_add_noise == 3) {
  set.seed(8316951 + 1000 * sim)
  # Add 3 noise
  x$noise1 <- rnorm(N, sd = 1)
  x$noise2 <- rnorm(N, sd = 1)
  x$noise3 <- rnorm(N, sd = 1)
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade3", "noise1",
    "noise2", "noise3")
}

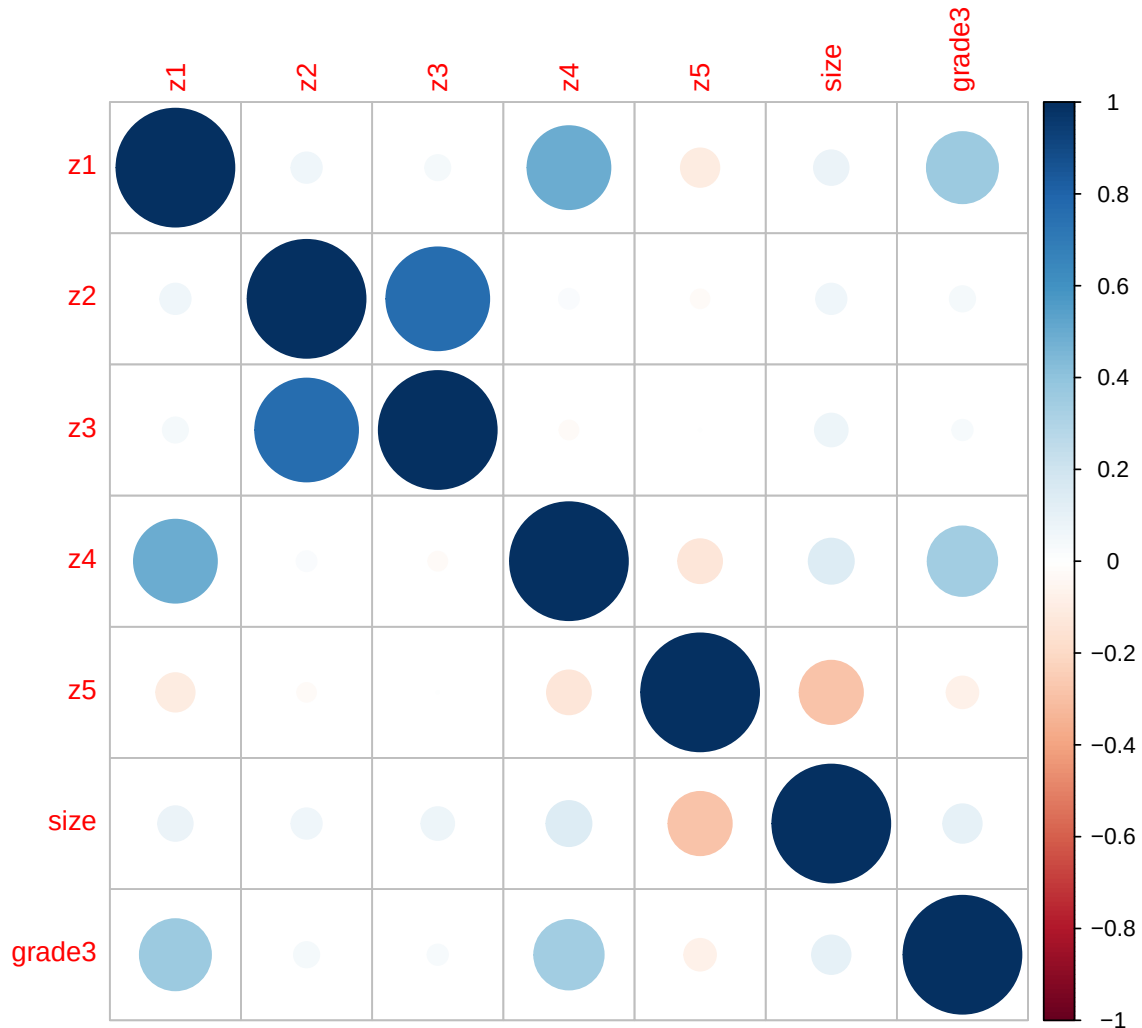
# More challenging which_replace <- which(confounders.name=='z1')
# confounders.name[which_replace] <- 'er'

cox.formula.check <- as.formula(paste("Surv(y.sim,event.sim)~treat+z1+z2+z3+z4+z5+size+grade3+noise1
coxph(cox.formula.check, data = x)

## Error in eval(predvars, data, env): object 'noise1' not found

Zm <- cor(as.matrix(x[, c(confounders.name)]))
corrplot(Zm)

```



```
suppressWarnings(table1(~z1 + z2 + z3 + z4 + z5 + size + grade3 + noise1 + noise2 +
  noise3 | treat, data = x))
```

```
## Error in eval(predvars, data, env): object 'noise1' not found
```

```
# Options Allconfounders.name is list of confounders within analysis dataset
# (1) use_lasso=TRUE & use_grf=FALSE Lasso used to possibly reduce dimension
# Any continuous factors are cut at medians (2) use_lasso=TRUE & use_grf=TRUE
# Lasso used to reduce dimension Continuous covariates are cut at medians
# However, if GRF selects a covariate cut then only that cut is used: For
# example if 'age <= median(ag)' is called for per Lasso but GRF includes 'age
# <= 54', then only the latter is used (3) use_grf_only = TRUE (overrides
# use_lasso and use_grf) Only factors selected via GRF are used (4) use_lasso =
# F & use_grf = T Median cuts (unless selected via GRF) as in (2) However no
# possible dimension reduction via lasso All categorical factors included (5)
# If all options set to false, then all factors are included with continuous
# factors cut at medians
```

```
use_lasso <- TRUE
```

```

use_grf <- TRUE
use_grf_only <- FALSE

fs.est <- forestsearch(df.analysis = x, Allconfounders.name = confounders.name, details = TRUE,
  use_lasso = use_lasso, use_grf = use_grf, use_grf_only = use_grf_only, dmin.grf = 12,
  frac.tau = 0.6, conf_force = NULL, outcome.name = outcome.name, treat.name = treat.name,
  event.name = event.name, id.name = id.name, n.min = nmin.fs, hr.threshold = hr.threshold,
  hr.consistency = hr.consistency, fs.splits = fs.splits, d0.min = d.min, d1.min = d.min,
  pstop_futile = pstop_futile, pconsistency.threshold = pconsistency.threshold,
  stop.threshold = stop.threshold, max.minutes = max.minutes, maxk = maxk, by.risk = 12,
  plot.sg = TRUE, vi.grf.min = vi.grf.min)

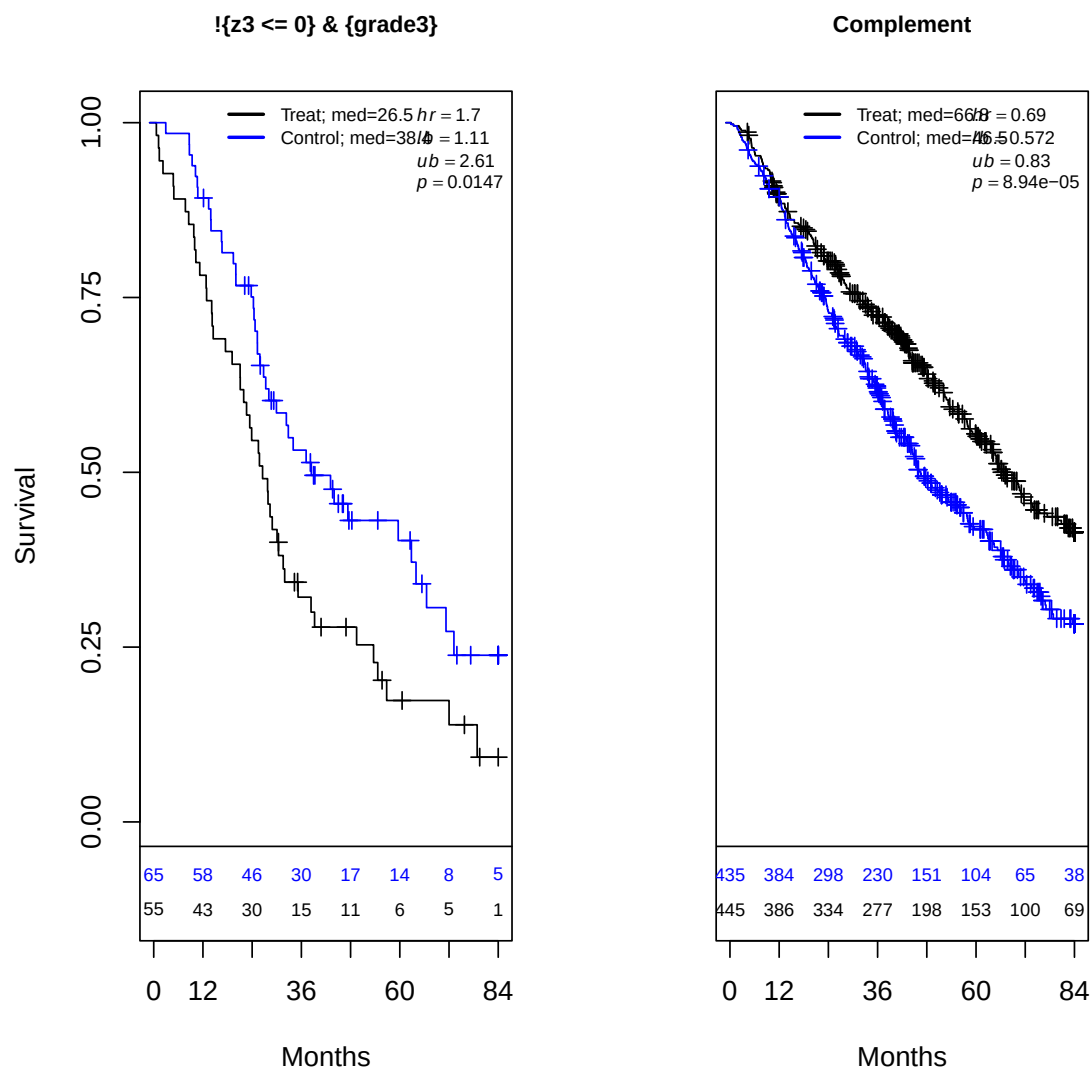
## tau, maxdepth= 50.1155 2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 1         2   -4.116917   577.000000   1.291563    4.116917   577.000000
## 2         3    1.017052   423.000000   1.370334   -1.017052   423.000000
## 3         4   -3.016608   656.000000   1.111991    3.016608   656.000000
## 4         5    1.392902    77.000000   3.565597   -1.392902    77.000000
## 5         6   -8.129389   144.000000   2.645706    8.129389   144.000000
## 6         7    8.918905   123.000000   2.881073   -8.918905   123.000000
##   treated.se      diff Nsg depth
## 1  1.291563  -8.233835 577     1
## 2  1.370334   2.034105 423     1
## 3  1.111991  -6.033216 656     2
## 4  3.565597   2.785803  77     2
## 5  2.645706 -16.258779 144     2
## 6  2.881073  17.837811 123     2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 6         7    8.918905   123.000000   2.881073   -8.918905   123.000000
##   treated.se      diff Nsg depth
## 6  2.881073  17.83781 123     2
## Subgroup found
## [1] "z1 <= 0"      "size <= 43" "z3 <= 0"
## [1] "z3 <= 0"
## # of continuous/categorical characteristics 1 6
## Continuous characteristics: size
## Categorical characteristics: z1 z2 z3 z4 z5 grade3
## CV lambda = 0.01410644
## 7 x 1 sparse Matrix of class "dgCMatrix"
##           s0
## z1      0.38231980
## z2     -0.02226776
## z3      .
## z4      0.48239404
## z5     -0.97595329
## size      .
## grade3  0.07264151
## Cox-LASSO selected: z1 z2 z4 z5 grade3
## Cox-LASSO not selected: z3 size
## Median cuts after Lasso
## Categorical after Lasso z1 z2 z4 z5 grade3
## Initial GRF cuts included z1 <= 0 size <= 43 z3 <= 0
## Factors included per GRF (not in lasso) size <= 43 z3 <= 0
## # of candidate subgroup factors= 7
## [1] "size <= 43" "z1"      "z2"      "z4"      "z5"
## [6] "grade3"     "z3 <= 0"

```

```

## Confounders per grf screening q5 q3 q7 q6 q2 q4 q1
## FSconfounders.name vi.cs vi_ratio
## 5 q5 0.1841072 1.2887507
## 3 q3 0.1816343 1.2714403
## 7 q7 0.1584832 1.1093826
## 6 q6 0.1310571 0.9174000
## 2 q2 0.1283427 0.8983990
## 4 q4 0.1264768 0.8853373
## 1 q1 0.0898986 0.6292902
## Number of unique levels (L) and possible subgroups= 14 16383
## # of subgroups based on # variables > k.max and excluded (per million) 0.016278
## k.max= 2
## Events criteria for control,exp= 10 10
## # of subgroups with events less than criteria: control, experimental 11 9
## # of subgroups with sample size less than criteria 16
## # of subgroups meeting all criteria = 89
## # of subgroups fitted (Cox model estimable) = 89
## *Subgroup Searching Minutes=* 0.005116667
## Number of subgroups meeting HR threshold 6
## # of candidate subgroups (meeting HR criteria) = 6
## Subgroups (1st 10) meeting overall screening thresholds (HR, m1) sorted by HRs= Inf
## K n E d1 m1 m0 HR L(HR) U(HR) q5.0 q5.1 q3.0 q3.1 q7.0 q7.1
## 1: 2 120 85 45 26.54 38.42 1.70 1.11 2.61 0 0 0 0 1 0
## 2: 2 123 97 49 21.07 30.53 1.61 1.08 2.40 0 0 0 0 1 0
## 3: 2 69 46 31 33.53 33.73 1.37 0.74 2.54 0 0 0 0 1 0
## 4: 2 155 102 53 30.07 43.11 1.35 0.91 1.99 0 0 0 1 0 0
## 5: 2 82 58 37 35.19 37.57 1.29 0.75 2.20 0 0 0 1 0 0
## 6: 2 159 117 59 28.36 32.82 1.26 0.88 1.82 0 0 0 1 0 0
## 7: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 8: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 9: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 10: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## q6.0 q6.1 q2.0 q2.1 q4.0 q4.1 q1.0 q1.1
## 1: 0 1 0 0 0 0 0 0
## 2: 0 0 0 1 0 0 0 0
## 3: 0 0 0 0 0 0 1 0
## 4: 0 1 0 0 0 0 0 0
## 5: 0 0 0 0 0 0 1 0
## 6: 0 0 0 1 0 0 0 0
## 7: NA NA NA NA NA NA NA NA
## 8: NA NA NA NA NA NA NA NA
## 9: NA NA NA NA NA NA NA NA
## 10: NA NA NA NA NA NA NA NA
## Consistency 0.99
## # of splits= 400
## Model, % Consistency Met= !{z3 <= 0} {grade3} 0.99
## Number of subgroups meeting consistency criteria=
## p.consistency Nsg group.id m.index K M.1 M.2
## 1: 0.99 120 1 1 2 !{z3 <= 0} {grade3}

```

```
## [1] "!{z3 <= 0}" "{grade3}"
## % consistency criteria met= 0.99
## SG focus= hr
## Subgroup Consistency Minutes= 0.02343333
## Subgroup found
## Minutes overall= 0.05498333

grf.est <- grf.subg.harm.survival(data = x, confounders.name = confounders.name,
  outcome.name = outcome.name, event.name = event.name, id.name = id.name, treat.name = treat.name,
  n.min = n.min, dmin.grf = dmin.grf, frac.tau = 0.6, details = TRUE)

## tau, maxdepth= 50.1155 2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 1         2   -4.116917   577.000000   1.291563    4.116917    577.000000
## 2         3    1.017052   423.000000   1.370334   -1.017052    423.000000
## 3         4   -3.016608   656.000000   1.111991    3.016608    656.000000
## 4         5    1.392902    77.000000   3.565597   -1.392902     77.000000
## 5         6   -8.129389   144.000000   2.645706    8.129389   144.000000
## 6         7    8.918905   123.000000   2.881073   -8.918905   123.000000
```

```
##   treated.se      diff Nsg depth
## 1   1.291563   -8.233835 577     1
## 2   1.370334    2.034105 423     1
## 3   1.111991   -6.033216 656     2
## 4   3.565597    2.785803  77     2
## 5   2.645706  -16.258779 144     2
## 6   2.881073   17.837811 123     2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 6         7      8.918905   123.000000   2.881073   -8.918905   123.000000
##   treated.se      diff Nsg depth
## 6   2.881073   17.83781 123     2
## Subgroup found
## [1] "z1 <= 0"      "size <= 43" "z3 <= 0"
## [1] "z3 <= 0"

# plot(grf.estftree)
```

```
library(doParallel)
registerDoParallel(parallel::detectCores(logical = FALSE))

cox.formula.boot <- cox.formula.sim
max.minutes <- 7
# Suggest running 50, first ... to get timing estimate

NB <- 1000

df_boot_analysis <- fs.est$df.est

fitH <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 0), cox.formula = cox.formula)
H_obs <- fitH$est_obs # log(hr) scale
seH_obs <- fitH$se_obs
# Hc observed estimates
fitHc <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 1), cox.formula = cox.formula)
Hc_obs <- fitHc$est_obs
seHc_obs <- fitHc$se_obs
rm("fitH", "fitHc")

Ystar_mat <- bootYstar({
  ystar <- get_Ystar(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
# Check dimension
if (dim(Ystar_mat)[1] != NB | dim(Ystar_mat)[2] != nrow(df_boot_analysis)) stop("Dimension of Ystar_")

tB.start <- proc.time()[3]
# Bootstraps
resB <- bootPar({
  ans <- fsboot_forparallel(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
tB.now <- proc.time()[3]
tB.min <- (tB.now - tB.start)/60

doParallel::stopImplicitCluster()

cat("Minutes for Boots", c(NB, tB.min), "\n")

## Minutes for Boots 1000 1.7243
```

```

cat("Projection per 1000", c(tB.min * (1000/NB)), "\n")
## Projection per 1000 1.7243

cat("Propn bootstrap subgroups found =", c(sum(!is.na(resB$H_biasadj_1))/NB), "\n")
## Propn bootstrap subgroups found = 0.667

# How many timed out
cat("Number timed out=", c(sum(is.na(resB$H_biasadj_1) & resB$tmmins_search > max.minutes)),
    "\n")
## Number timed out= 0

H_estimates <- get_dfRes(Hobs = H_obs, seHobs = seH_obs, H1_adj = resB$H_biasadj_2,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)
Hc_estimates <- get_dfRes(Hobs = Hc_obs, seHobs = seHc_obs, H1_adj = resB$Hc_biasadj_2,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)

print(H_estimates)

##           HO          sdHO HO_lower HO_upper          H1          sdH1 H1_lower H1_upper
## 1: 1.702336 0.3712214 1.110267 2.610137 1.436395 0.3295581 0.9161768 2.252001

print(Hc_estimates)

##           HO          sdHO HO_lower HO_upper          H1          sdH1 H1_lower
## 1: 0.6895097 0.06543284 0.572484 0.8304574 0.6888224 0.0934178 0.528041
##           H1_upper
## 1: 0.8985595

# H_estimates<-get_dfRes(Hobs=H_obs,seHobs=seH_obs,H1_adj=resB$H_biasadj_1,ystar=Ystar_mat,cov_metho
# Hc_estimates<-get_dfRes(Hobs=Hc_obs,seHobs=seHc_obs,H1_adj=resB$Hc_biasadj_1,ystar=Ystar_mat,cov_m
# print(H_estimates) print(Hc_estimates)

bootit <- list(H_estimates = H_estimates, Hc_estimates = Hc_estimates)
ans3 <- fsBoot.parallel.out(df = df_boot_analysis, FSboots = bootit, cox.formula.boot = cox.formula.
    sim_number = sim)
tall.min <- (tB.now - t.start.all)/60

cat("Overall minutes for analysis", c(tall.min), "\n")
## Overall minutes for analysis 1.902883

save(dgm, x, fs.est, grf.est, bootit, ans3, tall.min, resB, cox.formula.boot, file = file_out)

# This version is much slower than above Keep as place-holder but do NOT
# execute

library(doRNG)
library(doFuture)

registerDoFuture()
registerDoRNG()

plan("multisession", workers = 100)

cox.formula.boot <- cox.formula.sim

```

```

max.minutes <- 7
# Suggest running 50, first ... to get timing estimate

NB <- 100

df_boot_analysis <- fs.est$df.est

fitH <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 0), cox.formula = cox.formula)
H_obs <- fitH$est_obs # log(hr) scale
seH_obs <- fitH$se_obs
# Hc observed estimates
fitHc <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 1), cox.formula = cox.formula)
Hc_obs <- fitHc$est_obs
seHc_obs <- fitHc$se_obs
rm("fitH", "fitHc")

tB.start <- proc.time()[3]

Ystar_mat <- foreach(boot = seq_len(NB), .options.future = list(seed = TRUE), .combine = "rbind",
  .errorhandling = "pass") %dofuture% {
  ystar <- get_Ystar(boot)
}

cat("Done with Ystar_mat", "\n")
# Check dimension
if (dim(Ystar_mat)[1] != NB | dim(Ystar_mat)[2] != nrow(df_boot_analysis)) stop("Dimension of Ystar_mat")

cat("Starting FS bootstrap", "\n")

resB <- foreach(boot = seq_len(NB), .options.future = list(seed = TRUE), .combine = "rbind",
  .errorhandling = "pass") %dofuture% {
  ans <- fsboot_forparallel(boot)
}

tB.now <- proc.time()[3]
tB.min <- (tB.now - tB.start)/60

cat("Minutes for Boots", c(NB, tB.min), "\n")
cat("Projection per 1000", c(tB.min * (1000/NB)), "\n")

cat("Propn bootstrap subgroups found =", c(sum(!is.na(resB$H_biasadj_1))/NB), "\n")

# How many timed out
cat("Number timed out=", c(sum(is.na(resB$H_biasadj_1) & resB$tmins_search > max.minutes)),
  "\n")

H_estimates <- get_dfRes(Hobs = H_obs, seHobs = seH_obs, H1_adj = resB$H_biasadj_2,
  ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)
Hc_estimates <- get_dfRes(Hobs = Hc_obs, seHobs = seHc_obs, H1_adj = resB$Hc_biasadj_2,
  ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)

print(H_estimates)
print(Hc_estimates)

bootit <- list(H_estimates = H_estimates, Hc_estimates = Hc_estimates)

```

Table 1 Simulated data example: Cox hazard ratio (hr) estimates for the ITT population and subgroups H and H^c . Cox model estimates are based on subgroups: H true (knowing the actual subgroup, a-priori); the estimated subgroup $\hat{H}$; and the bootstrap (1,000 resamples) bias-correction to $\hat{H}$ estimates, denoted $\hat{H}_{bc}$. Estimates for the complement H^c are defined analogously. The number of subjects in each population (# Subjects) and the number of subjects correctly classified (# Correct) in the true subgroups H and H^c , respectively, are listed.

	Causal(adj)	Estimate	Lower	Upper	# Subjects	# Correct
ITT	0.806	0.77	0.649	0.913	1000	.
H subgroup estimates						
H_{oracle}	2.249	1.611	1.08	2.404	123	.
$\hat{H}$	1.536	1.702	1.11	2.609	120	68
$\hat{H}_{bc}$	1.536	1.436	0.915	2.253	120	68
H-complement subgroup estimates						
H_{oracle}^c	0.604	0.703	0.582	0.85	877	.
$\hat{H}^c$	0.707	0.69	0.574	0.83	880	825
$\hat{H}_{bc}^c$	0.707	0.689	0.529	0.898	880	825

```
ans3 <- fsBoot.parallel.out(df = df_boot_analysis, FSboots = bootit, cox.formula.boot = cox.formula.
  sim_number = sim)
tall.min <- (tB.now - t.start.all)/60
cat("Overall minutes for analysis", c(tall.min), "\n")

# save(dgm, x, fs.est, grf.est, bootit, ans3, tall.min, resB, cox.formula.boot,
# file=file_out)
```

```
## Confounders evaluated in Forestsearch:
## [1] "size <= 43" "z1"          "z2"          "z4"          "z5"
## [6] "grade3"     "z3 <= 0"
```